

FormulANN Grand Prix

Problem

k -nearest neighbor queries:

given a dataset S and a point q , find the k points in S closest to q

- $k = 100$
- 7 datasets
 - dimensions 384 - 3072
 - size 200k - 1.4M
- 1000 queries each
- query set
 - public for development
 - secret for the competition

Challenge prizes

Sherlock Holmes

fastest getting average recall ≥ 0.95

Bianconiglio

fastest getting average recall ≥ 0.8

Dory

approach using the least amount of memory with recall ≥ 0.95

Marie Kondo

fastest at building the index

Paperone

the one using the fewest distance computations

1st - 10pts

2nd - 8pts

3rd - 6pts

4th - 4pts

5th - 3pts

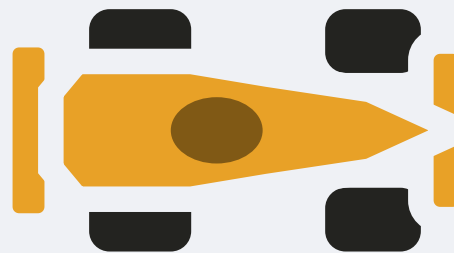
6th - 2ts

7th - 1pts

Neighbors grass



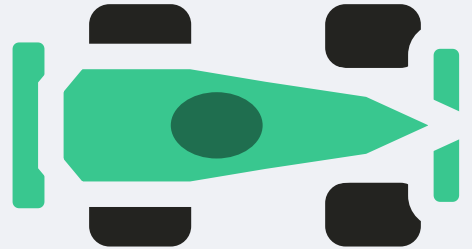
- Build HNSW graph on exact vectors
- Keep only RaBitQ quantized vectors for queries
- Never compute a full distance
- CPU with AVX512
- C++



Close Enough



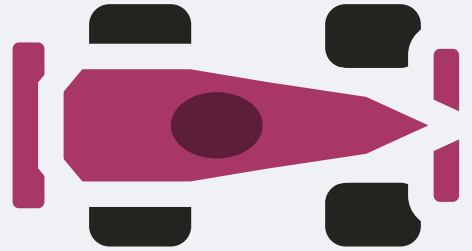
- Build HNSW graph on compressed RaBitQ vectors
- Tunes ef at build time, based on calibration queries
- Exact reranking
- CPU with AVX-512 instructions
- C++

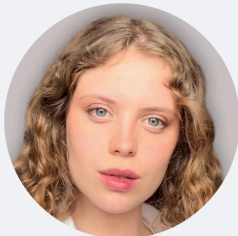


Kinda-neighbors



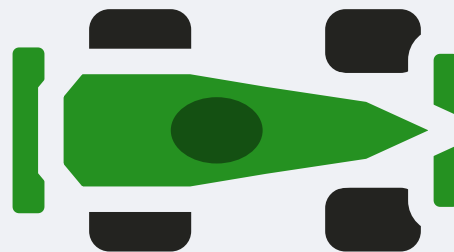
- Build HNSW graph on 8-bit quantized vectors
- Never computes a full distance
- Adds a patience parameter at the base HNSW layer
- Rust
- CPU

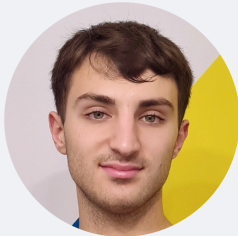
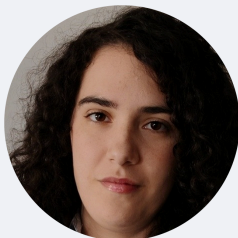




Friendly Neighborhood Solvers

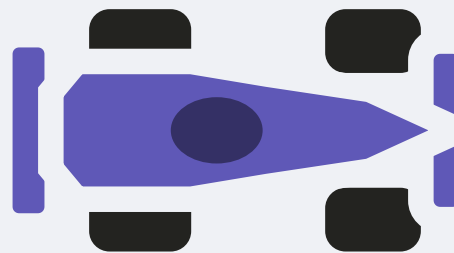
- Hierarchical k-means + quantization
- CPU for routing down the tree of clustering
- GPU for scoring vector codes
- C++





Finding NNemo

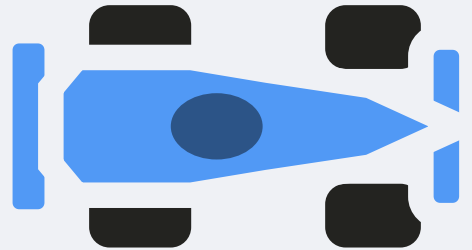
- IVF + SAQ quantization (SIGMOD26)
- No exact reranking
- Focus on minimizing memory footprint
- CPU
- C++



ANNarchy



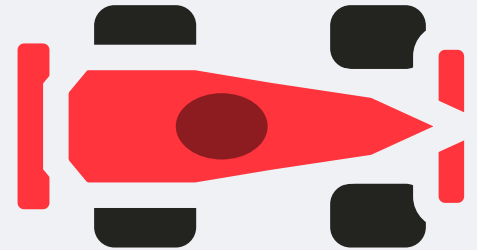
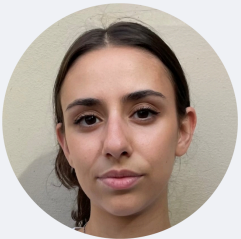
- HNSW or IVF, depending on the dataset
- No quantization
- CPU
- C++



ANNVedi



- GPU: Brute force over 8-bit quantized representation
- CPU: HNSW as a fallback for some datasets
- Verify only a limited number of candidates
- C++



Orthogonal Grand Prix

One circuit per dataset, driven by the per-query timings of the **high_recall** scenario.

Circuits

Paddock

Standings



agnews-mxbai-private

7 on the grid · 1000 queries



celeba-resnet-private

7 on the grid · 1000 queries



gooq-distilroberta-private

7 on the grid · 1000 queries



imagenet-clip-private

7 on the grid · 1000 queries



landmark-nomic-private

7 on the grid · 1000 queries



simplewiki-openai-private

7 on the grid · 1000 queries



yahoo-minilm-private

7 on the grid · 1000 queries

Conclusions

- Quantization matters, a lot!
- You can trust quantizer-estimated distances
- GPU is viable even in a query-at-a-time scenario